TECHNICAL PORTFOLIO | CONTROLS ENGINEERING

Three-Pump VFD Booster Control System

SIEMENS S7-1500 / TIA PORTAL V19 / WINCC RUNTIME ADVANCED / PLCSIM ADVANCED V8

PREPARED BY

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Software-based pressure booster plant with PI pressure regulation, variable-speed pump control, staged assistance and automatic standby-pump fault changeover.

4.0 bar	3 pumps	100 ms
CONTROL SETPOINT	SIMULATED VFD UNITS	PROCESS-CYCLE OB

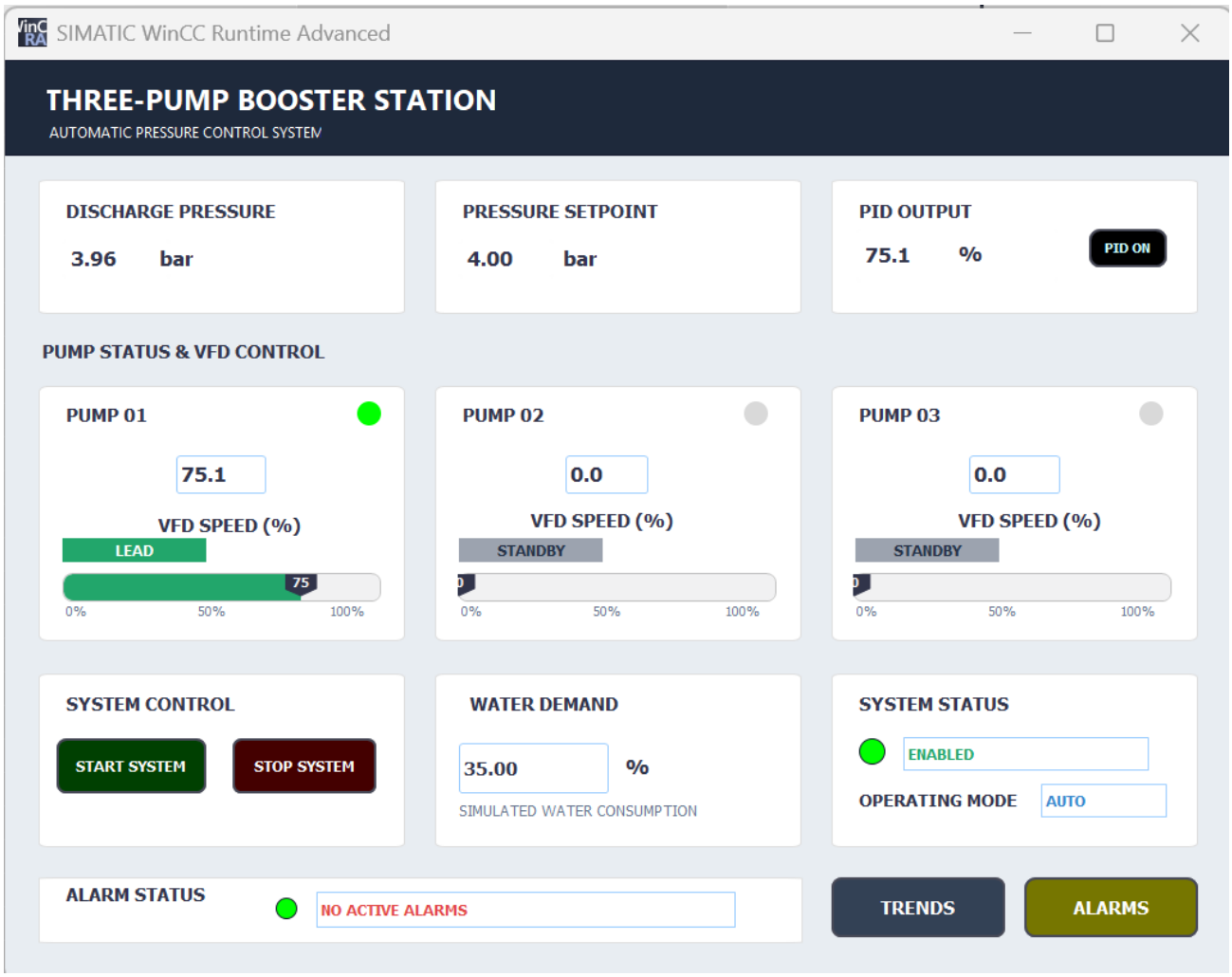


Figure 1. Normal operation at 35% simulated demand: Pump 1 is lead, discharge pressure 3.96 bar against a 4.00 bar setpoint.

Evidence basis | Screenshots supplied from the working WinCC Runtime simulation and recorded acceptance testing. No physical pumps, drives or sensors were commissioned.

01 | PROJECT INTENT

Plant architecture and operating scope

A modular Siemens PLC application regulates common discharge pressure while coordinating three simulated variable-frequency drives.

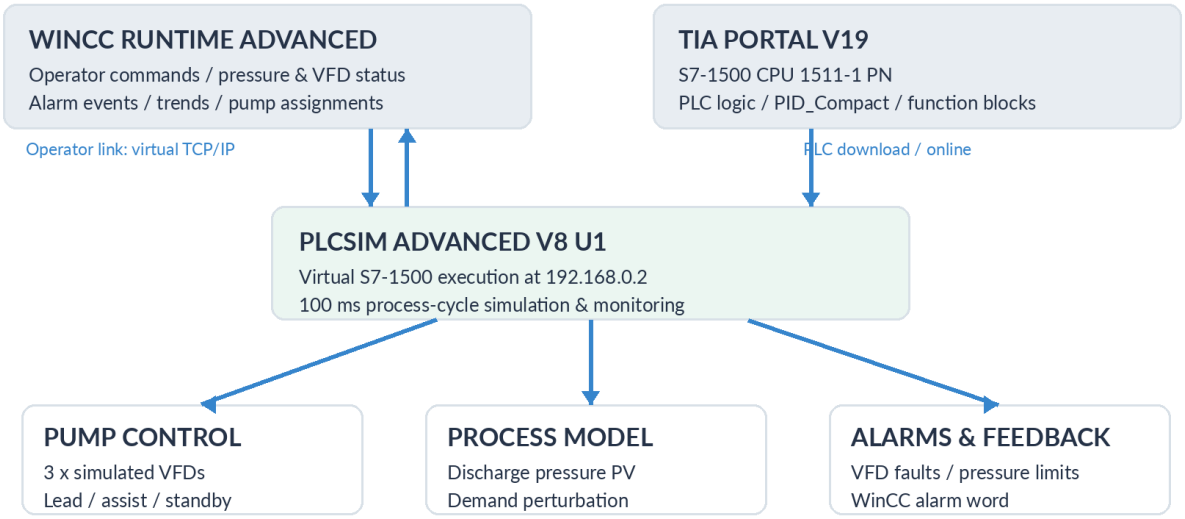


Figure 2. Logical control architecture. The diagram summarises the programmed and simulated components; it is not an electrical wiring drawing.

Software and simulated equipment

Component	Project configuration
Controller	Siemens S7-1500 CPU 1511-1 PN (simulated)
Engineering	TIA Portal V19; PLC logic in LAD and SCL
Runtime	WinCC Runtime Advanced; three HMI screens
PLC simulator	S7-PLCSIM Advanced V8 Update 1, virtual TCP/IP
Plant model	Three VFD speed responses, pressure PV and adjustable water demand

Design objectives

- Maintain a nominal 4.0 bar discharge-pressure setpoint under demand disturbances.
- Add or remove pumping capacity through delayed lead/assist staging.
- Transfer running duty to a healthy standby pump following a simulated VFD fault.
- Expose controls, process trends, equipment roles and operator alarms in WinCC.

02 | CONTROL ENGINEERING

PLC structure and control strategy

The implemented design separates process simulation, pressure regulation, pump assignment and alarm generation into reusable blocks.

Block / data	Responsibility
PID_Compact (PID_Pressure)	PI pressure loop: setpoint / process value / output in %
FB_PumpManager	Lead and assist allocation; staging delays; fault changeover
FB_PumpControl	Reusable run, speed request and fault handling per simulated VFD
FB_ProcessSimulation	Pump feedback dynamics and simplified hydraulic pressure response
FB_AlarmManager	Pump/pressure alarms and packed WinCC trigger word
DB_System / DB_Pumps	Operator commands, system signals and structured pump data

Pressure loop settings used for simulation

Parameter	Implemented value	Engineering interpretation
Setpoint	4.0 bar	Operator-adjustable in HMI
Control structure	PI	Derivative term disabled
Kp / Ti	10.0 / 5.0 s	Tuned against simulated process
Lead-pump speed	PID output, 0–100%	Applied to active lead VFD
Assist speed	30%	Fixed simulated assist reference

Sequencing rules

- Stage up when PID output ≥95% and pressure is more than 0.2 bar below setpoint, sustained for approximately 5 s.
- Stage down when PID output ≤50% and pressure has recovered to within 0.1 bar of setpoint, sustained for approximately 10 s.
- Healthy Pump 3 can assume lead or assist duty when another pump reports a simulated VFD fault.

These staging times derive from 50/100 counting cycles in the programmed 100 ms interrupt; actual elapsed delays depend on the entry conditions.

03 | CAPACITY STAGING

High-demand response: second pump assists

Increasing simulated demand requires additional hydraulic capacity. The lead VFD approaches high output before the lag pump enters service.

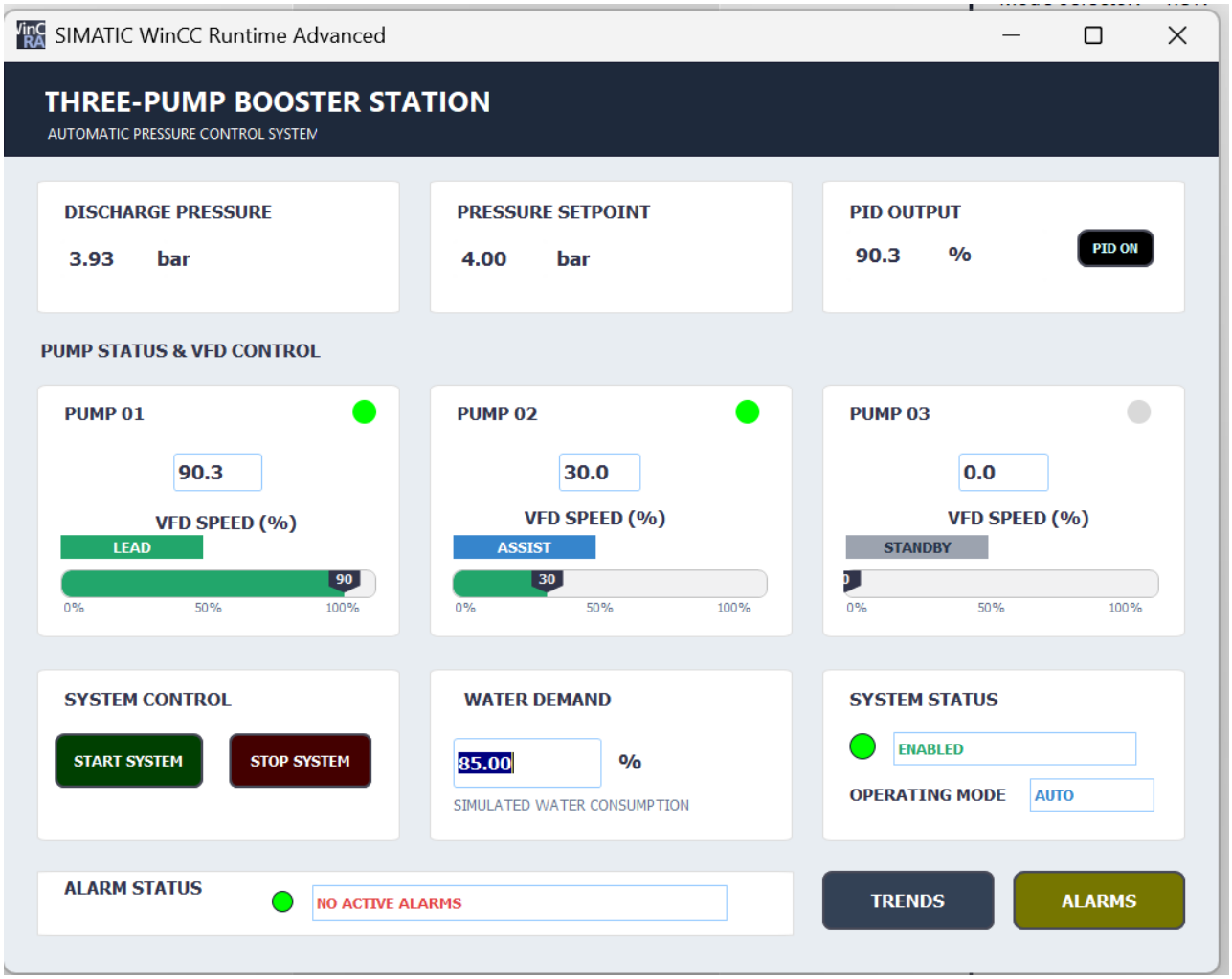


Figure 3. 85% demand capture: Pump 1 LEAD at 90.3%, Pump 2 ASSIST at 30.0%, pressure 3.93 bar versus a 4.00 bar setpoint.

Evidence point	Observed / recorded outcome
Demand step	35% → 85% from WinCC operator field
Automatic response	Pump 2 changed to ASSIST and ran at 30.0%
Pressure snapshot	3.93 bar at the time of the screenshot
Steady-state validation	4.00 bar achieved; user reported no sustained oscillation

RESULT

Additional capacity staged automatically, while the lead-pump PI loop retained discharge-pressure regulation.

04 | FAULT TOLERANCE

Pump 1 failure: standby lead takes over

The pump manager excludes the faulted drive and reassigns the PID-controlled lead role to a healthy pump.

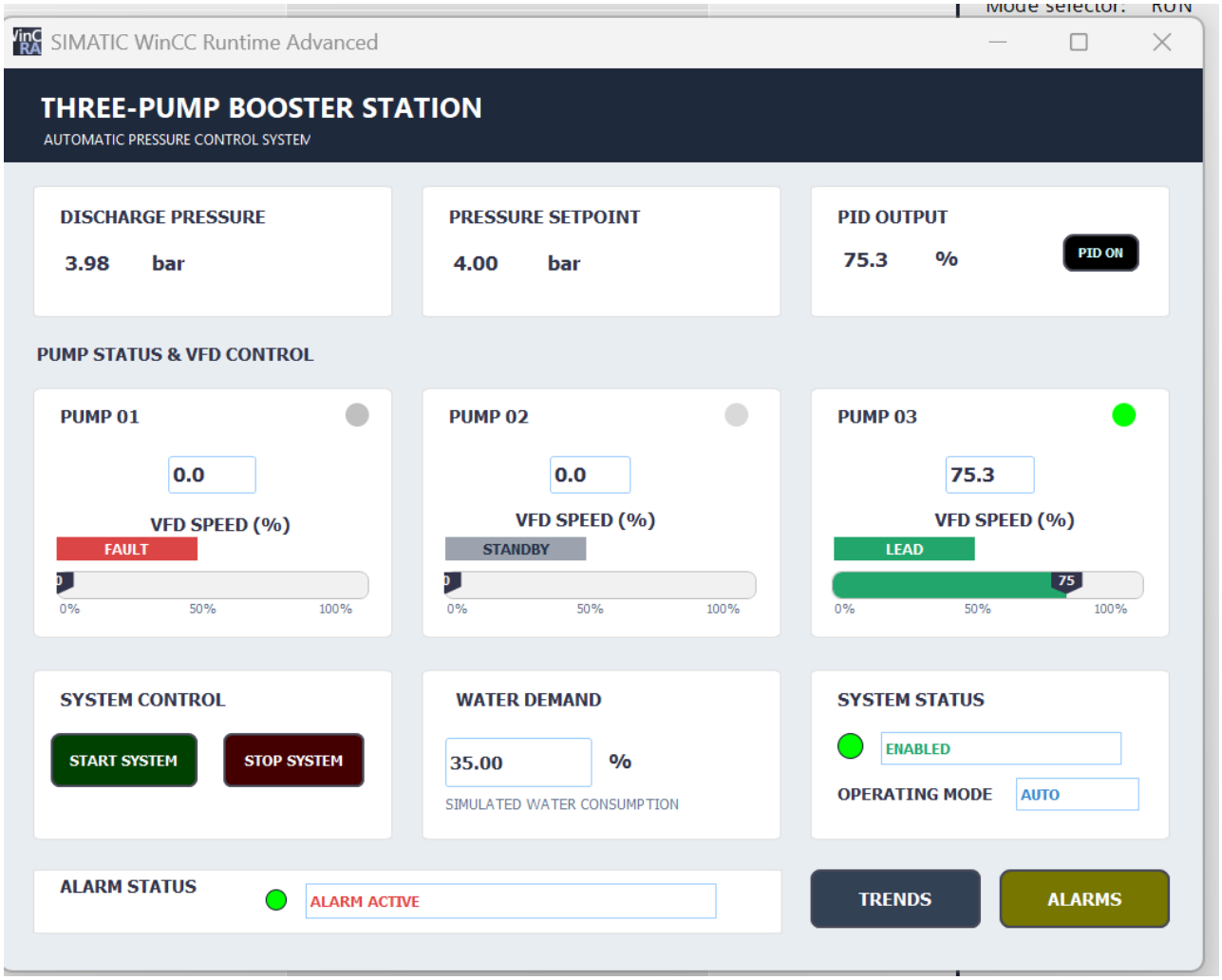


Figure 4. Pump 1 FAULT; Pump 3 LEAD at 75.3%; pressure 3.98 bar, close to the 4.00 bar setpoint.

Observed signal	Evidence from HMI
Pump 1	FAULT; speed 0.0%
Pump 3	LEAD; running at 75.3%
Pump 2	STANDBY; speed 0.0%
Discharge pressure	3.98 bar against 4.00 bar setpoint
Alarm status	ALARM ACTIVE

Changeover demonstrates basic simulated redundancy. It does not validate physical drive start-up, valve operation, electrical protection or fail-safe sequencing.

05 | FAULT TOLERANCE

Pump 2 failure: healthy assist selected

At elevated demand the control system can replace the faulted assist unit without abandoning pressure regulation.

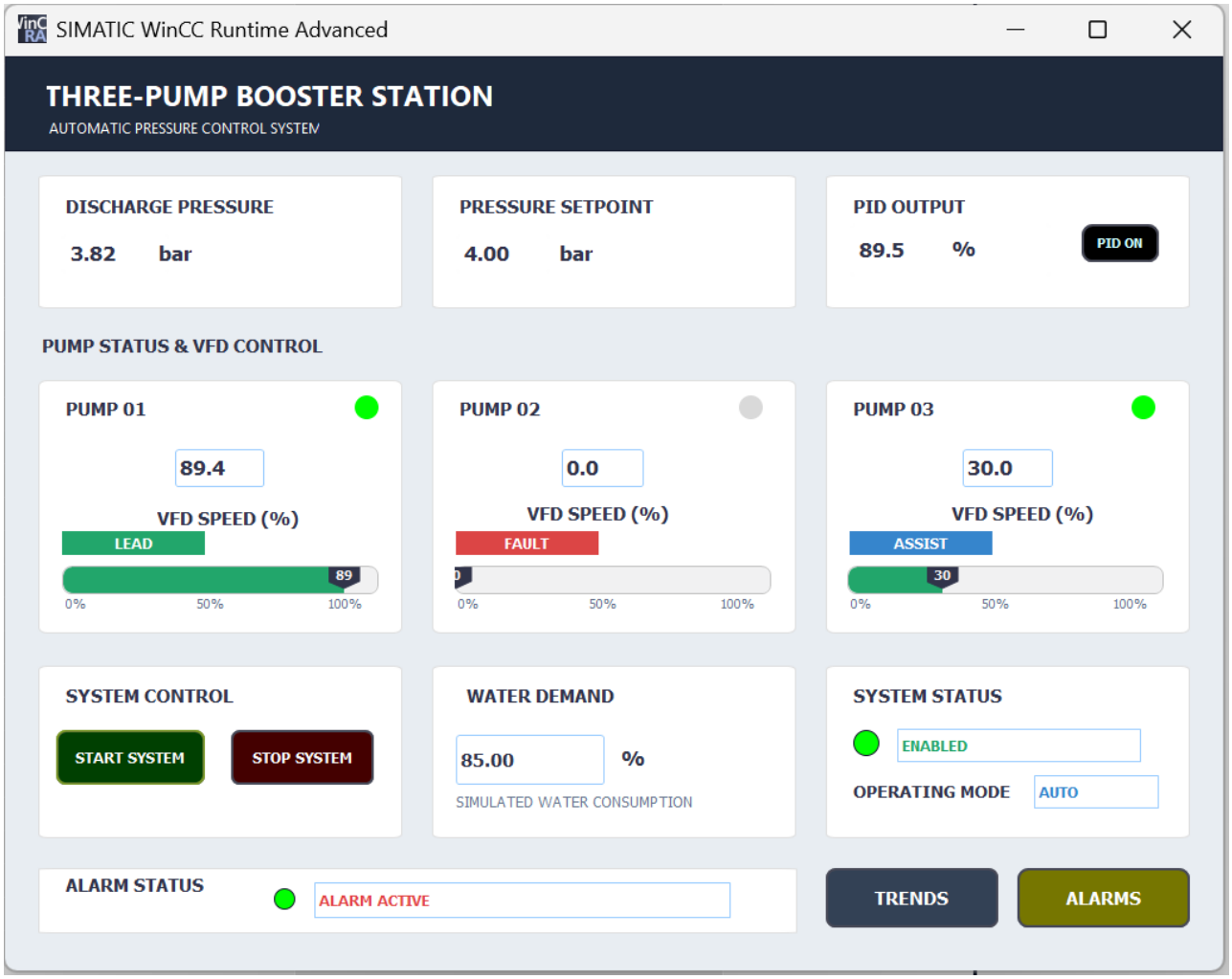


Figure 5. Pump 2 FAULT; Pump 1 LEAD at 89.4%; Pump 3 ASSIST at 30.0%; demand 85% and pressure 3.82 bar at capture.

Observed signal	Evidence from HMI
Demand / setpoint	85% / 4.00 bar
Pump 2	FAULT; speed 0.0%
Pump 1	LEAD; running at 89.4%
Pump 3	ASSIST; running at 30.0%
Discharge pressure	3.82 bar in the screenshot; subsequent recovery tested

The 3.82 bar value is a point-in-time observation, not a final steady-state error or a measured recovery duration.

06 | PROCESS DYNAMICS

PID response and pressure trending

The WinCC trend view displays actual pressure and target setpoint continuously, supporting disturbance-response diagnosis.

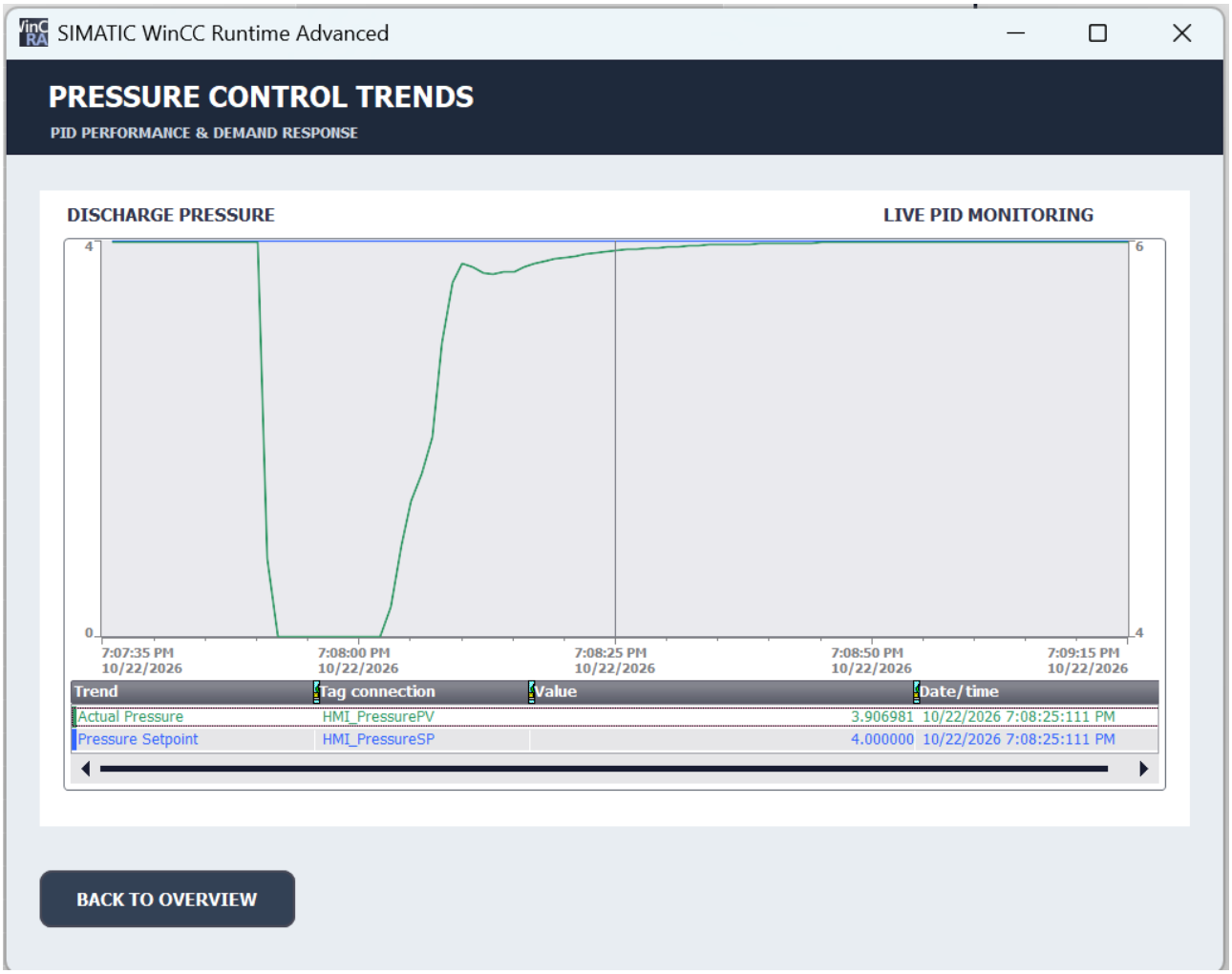


Figure 6. Pressure trend: a large transient followed by recovery of the measured pressure toward the flat 4.0 bar setpoint.

Engineering interpretation

- The pressure-process trace includes a deep drop toward zero before rising back toward the setpoint.
- The setpoint remains approximately 4.0 bar, showing closed-loop recovery after a simulated disturbance.
- The supplied single screenshot does not establish an exact settling time, peak overshoot or sampled error metric.

Improvement opportunity: add trend cursors or export logged samples so overshoot, rise time and settling time can be calculated quantitatively.

07 | OPERATOR DIAGNOSTICS

Discrete alarm detection and presentation

Pump VFD faults and high/low discharge-pressure warnings are converted to WinCC discrete alarms, with event timing and state information.

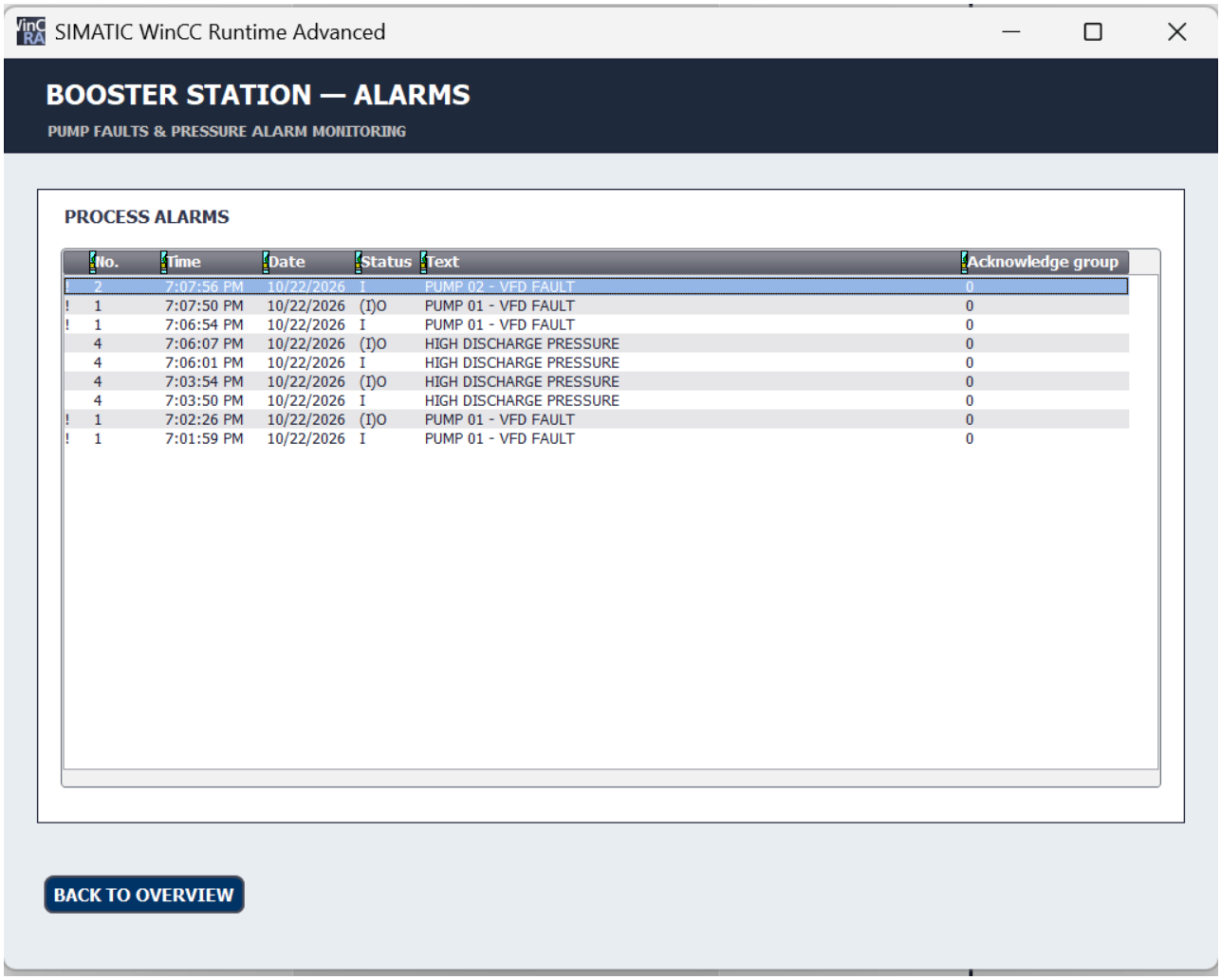


Figure 7. WinCC alarms view shows Pump 1/Pump 2 VFD-fault events and high-discharge-pressure entries with timestamps.

Alarm class	PLC data source / WinCC mapping
Pump 1 / 2 / 3 VFD fault	DB_System pump fault alarms → AlarmWord bits 0–2
High / low pressure	Pressure alarms → AlarmWord bits 3–4
Overview summary	DB_System.AnyAlarm → ALARM ACTIVE / NO ACTIVE ALARMS

Event display and time stamps are evidenced. Long-term alarm archival and physical safety functions were not demonstrated in the supplied captures.

08 | CAPACITY REDUCTION

Demand reduction and pump stage-down

With demand reduced to 20%, the assist unit is removed from service and Pump 1 continues as the pressure-regulating lead pump.

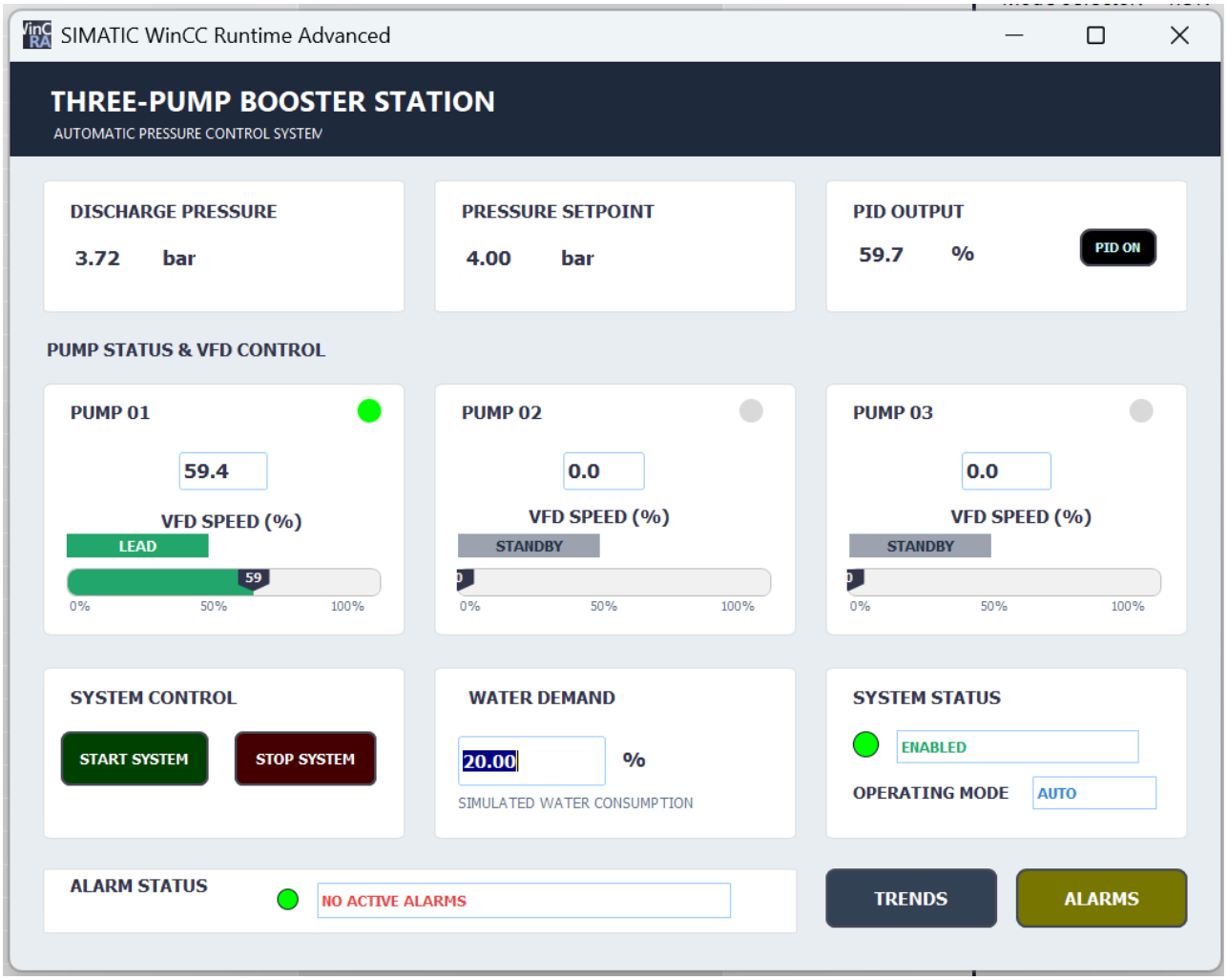


Figure 8. 20% demand capture: Pump 1 LEAD at 59.4%, Pumps 2 and 3 STANDBY. Pressure reads 3.72 bar during the captured response.

Check	Outcome
Demand command	Reduced from 85% to 20%
Pump assignment	Only Pump 1 remains LEAD
Pump speeds	Pump 1 ≈59.4%; Pumps 2 / 3 at 0%
Pressure recovery	User subsequently confirmed the stage-down and stabilisation test passed

The screenshot is a transient capture (3.72 bar), not proof that the final pressure was exactly 4.0 bar at that instant.

09 | VALIDATION SUMMARY

Acceptance evidence and engineering boundaries

The project has been functionally exercised in simulation. The matrix below distinguishes observed screenshots from previously reported acceptance results.

Test / function	Outcome	Evidence
Normal pressure control	Pressure 3.96 bar at 35% demand	Figure 1
High-demand assistance	Pump 2 assisted; final 4.00 bar reported	Figure 3 + test notes
Stage-down at 20%	Pump 2 stopped; recovery validated	Figure 8 + test notes
Pump 1 fault changeover	Pump 3 became lead; pressure 3.98 bar	Figure 4
Pump 2 fault changeover	Pump 3 became assist; pressure 3.82 bar snapshot	Figure 5
Trending	PV response and setpoint visible	Figure 6
WinCC alarms	Time-stamped process alarm events visible	Figure 7
PLC / WinCC integration	Live tags, operator commands and indicators tested	Figures 1, 3–5

Boundaries of the validation

- Software simulation only: no physical S7-1500 I/O, pumps, VFDs, pressure transmitter or hydraulic pipework.
- Drive interlocks, emergency stop, protective trips, network recovery and cybersecurity were not independently tested on hardware.
- PID gains and staging thresholds are model-specific; physical commissioning would require plant identification and retuning.
- Screenshots are operational evidence, not a substitute for reviewed PLC source code or timestamped measurement logs.

PORTFOLIO POSITIONING

Demonstrates S7-1500 control design, PI regulation, VFD simulation, lead/assist sequencing, fault handling, WinCC HMI engineering and structured functional validation.